

Which transfer strategy survives leaving the laboratory?

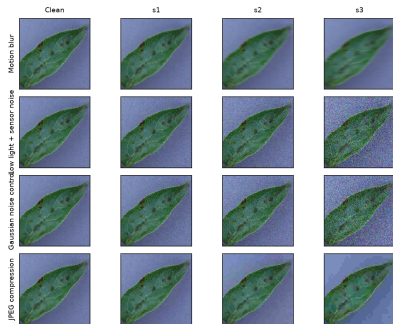
PlantVillage: 38 plant diseases, photographed in a lab — one leaf, grey background, even light. A pretrained ResNet-50 saturates on it.

So we ask a harder question. Same network, **one variable**: which parameters get gradients.

- **Linear probe** — backbone frozen, 78 k parameters
- **Fine-tuned** — all 23.6 M parameters

Then we photograph the test set with a **field robot** instead: motion blur, low light with sensor noise, JPEG for the radio link.

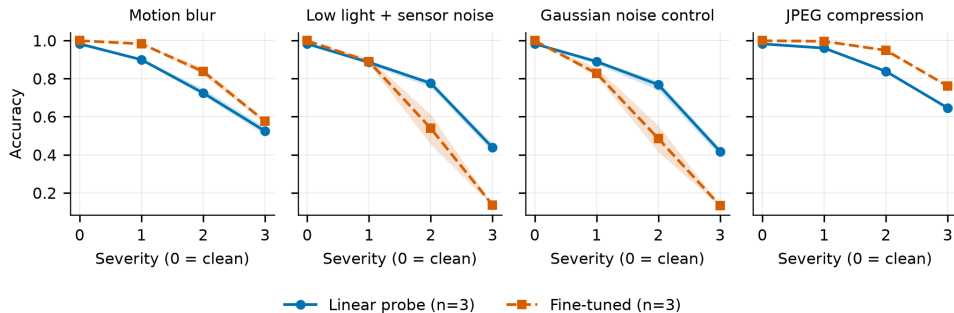
Three severities, three seeds, no retraining.



Rows: blur, low light, Gaussian control, JPEG.

Columns: clean → severity 3.

Fine-tuning wins — until the noise starts



Clean data, seed 42: **3 errors against 34.**

Blur and JPEG: fine-tuning ahead everywhere, by up to 11 points.

Sensor noise: **the order reverses**, by 24 points at severity 2.

Every seed agrees — the best fine-tuned run still loses to the worst probe.

Why, what breaks, and what to take away

A control narrows the cause.

The third panel is pure Gaussian noise, same brightness, no gain. It reproduces the reversal, one severity earlier.

So noise alone is enough. *Why* it hurts fine-tuning — filters tuned to lab texture — is still a hypothesis, not a result.

The failure has a shape.

Under noise the fine-tuned model sends **61–76 % of its errors into one class** — Septoria leaf spot, which looks like dark speckle. The probe drifts there too, far less.

And it stops admitting doubt.

Clean: fine-tuning is the better calibrated. Low light, severity 2: ECE **0.26–0.42** against **0.17–0.20**, three seeds.

Take away.

- Fine-tune by default.
- Robustness is not one number — it flips with the *kind* of degradation.
- Report calibration, not only accuracy.